Toward Universal Representation Learning for Multidomain Hyperspectral Image Classification

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***Abstract*— Deep learning-based methods have greatly improved the performance of hyperspectral image (HSI) classification over the past several years. Nevertheless, current deep learning methods require training and deploying an independent model for each hyperspectral data domain. Representations learned for one data domain can hardly be generalized to other data domains, so multiple models are needed in real-world applications when data from multiple domains are involved. In this article, we design a single neural network that learns universal representations simultaneously from multiple hyperspectral remote sensing data domains. The universal convolutional neural network (CNN) adapts its behavior to different hyperspectral datasets. The majority of parameters of the network are shared to learn common knowledge from multiple datasets. A small number of domain-specific parameters are assigned to handle the domain shift. In addition, we propose a two-step training strategy to fully utilize the capacity of the universal network. Experiments conducted on seven HSI datasets demonstrate that the proposed universal network outperforms multiple individual specialized single-domain networks.**

***Index Terms*— Hyperspectral image (HSI) classification, multidomain learning, neural networks, universal representation.**

I. Introduction

COMPARED to panchromatic or color imaging, hyperspectral imaging technology acquires the reflectance of target objects in a broad, continuous spectrum of light wavelength. Each pixel in obtained hyperspectral images (HSIs) is a vector depicting responses at hundreds of hyperspectral bands, making HSI particularly to be powerful in distinguishing objects made of different materials. Therefore, HSI has been used in many remote sensing applications [1], such as agriculture [2], environment [3], and urban planning [4].

Image classification is a fundamental research problem of hyperspectral remote sensing. The goal is to assign a label to each pixel of HSI. This problem has been actively studied during the last several decades. Researchers in the remote sensing community have proposed many HSI classification algorithms, from traditional shallow methods to deep learning-based methods [5], [6], [7], [8].

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With carefully designed network architectures and optimization methods, the classification accuracy of hyperspectral remote sensing images has been increasing quickly. On certain datasets, existing methods from the literature achieved near 100% classification accuracy [9]. However, most methods have significant limitations due to the fact that they are highly specialized in one data domain. In the current paradigm, we need to design and train a new model for each new dataset. In real-world applications, this paradigm leads to significant inconvenience and disadvantages.

From model design and training perspectives, it takes a significant amount of expert knowledge and effort to redesign a new model for every new dataset. In addition, retraining each new model requires a lot of computational resources and time because of the high cost of the computational load of training deep neural networks. From the application perspective, the task and data modality of interest may be constantly changing. To deploy deep learning models in real-world applications, the current single model paradigm requires a large number of domain-specific trained models to be available on the image processing device. This increases the burden of storing and transmitting large deep learning models [10]. Moreover, frequently shifting and reloading models with numerous parameters lead to challenges in application scenarios [11]. This issue becomes especially severe when models are deployed on airborne or spaceborne platforms. Therefore, a single neural network that can simultaneously work on multiple hyperspectral datasets, which can be considered a universal model, is needed to address the above challenges. In this article, we investigate the problem of universal representation learning for hyperspectral remote sensing image classification in the framework of multidomain learning.

Many machine learning algorithms assume that all data points in the dataset follow independent identical distribution [12]. If a model is trained on a dataset with one distribution, it usually does not perform well on a different dataset with another distribution. Careful design and adaptation need to be taken into account in the multidomain scenario. Hyperspectral remote sensing images differ significantly in data modality and distribution in many different ways. First, different hyperspectral sensors have different designs and produce different spectral responses to electromagnetic waves. For example, the AVIRIS sensor responds to a bandwidth of 0.4–2.5 μm, while the ROSIS sensor has a spectral response range of 0.43–0.86 μm. This means that the physical meanings of different spectral bands are different. Second, different hyperspectral sensors have different spectral dimensions. Thus,

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the collected HSIs have different sizes in the spectral domain. This is different from natural image classification where the input image has one band for a binary or grayscale image or three bands for a color image. Third, even the datasets captured by the same hyperspectral sensor may have different data distributions. For instance, an image collected over a rural area should have a different distribution from an image collected over an urban area. For the same imaging area, hyperspectral remote sensing is also easily affected by weather, season, illumination, and other factors. In addition, the number of classes and the semantics of classes considered in each dataset may also be different. Therefore, the collected HSIs have different data distributions under different scenarios.

All of the above-described domain and data modality shifts of hyperspectral remote sensing images make it challenging for a single model to classify multiple hyperspectral datasets. However, the human visual system has the ability to solve a large number of image recognition tasks with different data distributions [13]. We can recognize a wide range of natural images containing various natural scenes and man-made objects. Images with significantly different distributions, such as sketches, paintings, and medical images, can also be understood and classified by humans. This suggests that universal representations of images may exist in the human visual system. Thus, it is valuable to explore the topic of universal representations of HSIs with computational algorithms. Here, we define multidomain learning in HSI as learning a universal representation by a single model on multiple HSI datasets simultaneously. Different from domain adaptation that optimizes performance on the target domain and often forgets the source domain, the aim of multidomain learning is to obtain a universal model/representation that performs the best on all domains. It also differs from multitask learning that performs multiple tasks on a single domain. The recent endeavor in the computer vision community also suggests that the universal model is a promising direction and a step toward general artificial intelligence [10], [11], [13], [14], [15].

Lee et al. [16] proposed a method to train multiple hyperspectral datasets at the same time. Their results showed a certain benefit of this training scheme. Later, the authors extended their work to answer the question of whether pretraining is necessary for HSI classification [17]. Multiple datasets were used to pretrain a network, and then, this network was used to fine-tune another small dataset. The main drawback of these two methods is that they simply feed multiple datasets to one network but did not take the dataset domain discrepancy into consideration.

Overall, despite achieving significant success in HSI classification, traditional deep learning paradigms still face several challenging issues due to the single-domain training protocol. First, it requires designing and training a new model for each new dataset, which is very inconvenient. Second, parameters and knowledge learned on one dataset cannot be shared and reused on other datasets, making it impossible to fully utilize the potential of multiple data domains. Third, deploying one model for each dataset in the real application requires large storage and memory to store and run multiple models. These issues motivate us to develop a universal representation learning method on multiple hyperspectral datasets. In addition, the large diversity and shifts among hyperspectral domains motivate us to design modules to address the domain shifts by domain-specific parameters.

In this article, we propose a universal network for multidomain HSI classification (MHIC). To achieve universal representation learning, the single network should be trained on multiple HSI datasets. In the deployment stage, only a single universal model needs to be deployed to make predictions on multiple datasets. The universal network tries to explore common knowledge of different HSIs. At the same time, a small set of domain-specific modules is employed to compensate for the specific property of each dataset. This enables the universal network to be optimized on multiple datasets simultaneously. In specific, we have designed adaptive modules for addressing domain shifts and discrepancies of multiple datasets so that different datasets do not conflict with each other in the training. Furthermore, we designed a two-stage training strategy to train the universal network to fully optimize it. We evaluate the effectiveness of the adapters in different convolutional neural networks (CNNs) and investigate the locations to insert the adapters along with layers in a deep network. In addition, when new datasets are available in the future, the universal network is easy to generalize to new data without the need to retrain the whole network.

Compared to existing methods, our proposed multidomain universal network has the following advantages. First, it is a single network capable of learning on multiple hyperspectral datasets simultaneously. Thus, the network can fully utilize common knowledge across multiple datasets. The traditional network only learns on one dataset and fails to use more data. Second, in real applications, the universal network allows to deploy only one network for multiple datasets. This avoids storing many models and an excessive number of parameters in memory. Third, the proposed network can be easily generalized to new datasets by only adding and training an adapter with a very small number of parameters. At the same time, all knowledge learned from the previous datasets is kept without loss.

The contribution of this article can be summarized as follows. First, we propose a universal representation learning network for MHIC. This network can learn common representations shared by different hyperspectral datasets or domains. Second, we design adapter modules to compensate for domain shifts when training the network on multiple datasets. This is the first time that the domain shift problem is addressed in HSI multidomain learning. Third, our proposed single network can achieve similar or better classification performance than multiple models trained for an individual domain with significantly reduced model size.

The rest of this article is organized as follows. Section II reviews works related to HSI classification and multidomain learning. Section III provides the details of the proposed universal network model and the strategy to address the domain shift problem. Section IV demonstrates comprehensive experiments to evaluate the proposed design on seven HSI datasets. Section V concludes this article.

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II. Related Work

In this section, we first briefly review single-domain HSI classification methods. Then, we explain the topic of multidomain learning and its differences from other similar research topics.

A. Single-Domain HSI Classification

Most HSI classification methods can be divided into two groups: traditional methods and deep learning-based methods. Traditional methods usually run feature extraction or dimensionality reduction first. Then, the extracted low-dimensional features are classified using shallow classifiers, such as support vector machine (SVM) and random forest. Liao et al. [18] proposed to use local discriminant analysis to extract features from HSI and then SVM is used to predict the final labels. Yang et al. [19] proposed a nonparametric hyperspectral feature extraction method, which is combined with the nearest neighbor classifier for classification. Surveys of traditional HSI classification methods can be found in [20], [21], and [22].

Due to the development of parallel computation and advances in network design, deep learning has become the state-of-the-art paradigm for hyperspectral remote sensing image classification, achieving superior results on most hyperspectral benchmark datasets. A variety of deep neural networks have been adapted for HSI classification in the last few years. Typical models include deep belief network [23], stacked autoencoder [24], deep recurrent network [25], and deep CNN [7], [9], [26], [27], [28]. Of all the deep learning models, the deep CNN has obtained the most attention. CNN can extract discriminative spectral, spatial, or spectral–spatial features using 1-D, 2-D, or 3-D convolutions [7]. These features are then classified by a Softmax classifier. All parameters in most CNNs are trained end-to-end on a single-domain hyperspectral dataset each time.

B. Multidomain Learning

The goal of MHIC is to design a single universal model that can be trained and deployed on multiple datasets simultaneously. To achieve this, the solution should be able to extract universal representations for HSI. This new topic has not been well addressed in the literature. Here, we review some research topics related to multidomain universal representation learning and compare them with our problem.

Multitask learning is closely related to multidomain learning. The main aim of multitask learning is to design a single model to simultaneously solve different tasks. The basic assumption behind multitask learning is that different tasks have complementary information so that each task can benefit from the training of other tasks. In the context of deep learning, one solution is to share shallower layers of neural networks for all tasks and use task-specific parameters in deeper layers for individual tasks [29]. In computer vision, Ranjan et al. [30] proposed a method to simultaneously perform landmark localization, face detection, and pose estimation. Other multitask image applications have also been explored [31], [32], [33]. The main difference between multitask learning and multidomain learning is that the former learns on a single data domain for multiple tasks. However, multidomain learning learns on different data domains for a single task.

Transfer learning and domain adaptation also have a close relationship to multidomain learning. Transfer learning aims to transfer knowledge learned from one dataset to another dataset. In deep learning, the paradigm of pretraining on Imagenet and fine-tuning on another dataset is such an example [34]. When the label on the target dataset is not available, unsupervised domain adaptation is usually used [35], [36], [37], [38]. In hyperspectral classification, an active transfer learning method is proposed to achieve good performance using only minimally labeled training data [39]. Yang et al. [40] proposed a heterogeneous graph attention network for unsupervised multiple-target domain adaptation. The main difference between transfer learning from multidomain learning is that the goal of transfer learning is to optimize the performance of the target dataset. The performance on the source dataset is normally degraded after transfer learning. Multidomain learning, on the contrary, tries to maximize the performance on both source and target datasets.

With the release of the Visual Decathlon challenge [41], the topic of multidomain learning has drawn attention in the computer vision community. The Visual Decathlon is a challenge that learns a single model for classification on ten different datasets. Researchers have proposed a series of methods for this challenge [11], [13], [42], [43], [44]. Following this, Wang et al. [10] designed a challenge for multidomain object detection. Recently, the universal representation has been investigated in few-shot scenario [14], [15], [45], [46], [47]. Multidomain representation has also been studied in medical imaging area [48], [49]. However, the multidomain learning issue has not been addressed in the HSI classification. Lee et al. [16] considered a similar problem, but their method mainly follows the single-domain learning design and does not have a module customized for solving issues in multidomain learning. Our work is the first attempt to formally define and design modules to address the domain shifts and learn universal representations in multidomain learning for HSI classification.

III. Proposed Method

In this section, we describe the proposed universal representation learning method in detail. First, we formally define the problem of universal representation learning on multiple domains. Then, we illustrate the overall framework of our solutions. Finally, the designed adapters for multidomain hyperspectral classification are described.

A. Problem Formulation

The goal of multidomain learning for universal representation is to design a single model that can be trained and tested on hyperspectral datasets from different domains. Formally, suppose that we have T datasets $\{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_T\}$, where $\mathcal{D}_t$, $t = 1, 2, \ldots, T$, represents one dataset. We use dataset and domain interchangeably in this article. Each dataset $\mathcal{D}_t$ is composed of a pair $\{X_t, Y_t\}$. $X_t = \{x_i^t\}$, $i = 1, 2, \ldots, N_t$, where N_t denotes the number of training samples in $\mathcal{D}_t$, $x_i^t \in \mathcal{R}^{A \times A \times U_t}$

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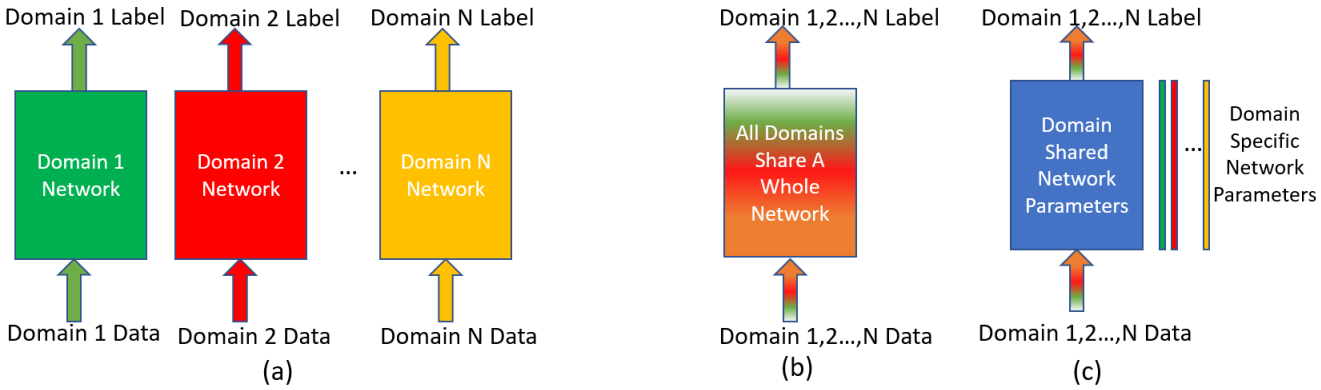


Fig. 1. Illustration of different schemes to achieve multidomain learning. (a) Multiple single models. This is the traditional paradigm to train an individual network for each data domain. Different colors represent different data domains and corresponding networks. (b) Single shared model. All data domains share all network parameters in a single network. The arrows and box with multiple colors represent multiple data domains and the network that processes all the data. (c) Proposed multidomain model with adapters. The network consists of a part shared by all data domains and domain-specific modules.

denotes a hyperspectral sample in dataset $\mathcal{D}_t$, and A and U_t are the spatial size of patches and number of spectral bands in the tth dataset. $Y_t = \{y_i^t\}$, where $i = 1, 2, \ldots, N_t$ is the label of the ith sample in $\mathcal{D}_t$, $y_i^t \in \{1, 2, \ldots, V_t\}$ represents one label in $\mathcal{D}_t$, and V_t is the number of categories. Let F denote a neural network model; $\mathbf{W}$ represents its parameters, and l represents the loss function. We can obtain the objective function for a single domain $\mathcal{D}_t$ as

$$L_t = \frac{1}{N_t} \sum_{i=1}^{N_t} l\left(\mathsf{F}\left(x_i^t; \mathbf{W}\right), y_i^t\right). \tag{1}$$

To achieve good performance on multiple datasets, we need to consider the losses on all datasets. There are some possible ways of balancing performances across multiple datasets. To make it simple and following [12], we opt to use the average of losses on all datasets as our final objective function

$$L_{\text{all}} = \frac{1}{T} \sum_{t=1}^{T} L_t \tag{2}$$

where L_t is defined in (1).

B. Overall Framework

Several potential approaches are available for multidomain learning, as illustrated in Fig. 1. In the traditional paradigm of deep learning, an individual model $\mathsf{F}_t(; \mathbf{W}_t)$ has to be trained for each dataset $\mathcal{D}_t$. Then, a bank of individual models is obtained and deployed for multidomain image classification. In this case, the total number of parameters and size of models will increase very quickly with the number of domains because the total model size is the sum of all individual models. The representations are learned for each domain specifically, so there are no universal representations in this solution. This scheme is illustrated in Fig. 1(a), where each color represents a data domain and its network. To reduce the number of parameters for all domains, a naive solution is to simply make all datasets share one common network, as shown in Fig. 1(b). Multiple colors in the box denote that multiple data domains share the same network. The obtained representation can be considered a universal representation for all domains. However, this will cause learning difficulties for the network. Since different data domains have different distributions and statistics, plain machine learning models normally experience performance degradation when facing datasets with different distributions [50], [51]. Although some methods, such as batch normalization (BN) and domain adaption, have been proposed to address this issue, they are not designed for multidomain scenarios. Thus, simply sharing the same network for all datasets is not an optimal solution.

Our proposed scheme is shown in Fig. 1(c). To address the domain shift issue in multidomain classification, our proposed multidomain network (MDN) is equipped with modules to adapt the discrepancy of different distributions in multiple datasets. Generally, a base network shared by all data domains is used to extract features from all datasets. The extracted features work as basic features for multidomain classification. It is assumed that, although different HSI datasets have different attributes, there are certain common properties shared by multiple hyperspectral datasets. The base network is supposed to learn such shared feature representations. Thus, the parameters of the base network are shared, and the computation and optimization of the base network are performed on all available training domains. The blue box in Fig. 1(c) represents the shared base network.

However, the features extracted by the base network may not be optimal for any dataset because they are optimized on all datasets rather than a single one, given the differences in data distributions of all datasets. Therefore, we designed adapter modules to adjust the basic features for each dataset, as shown in Fig. 1(c). Each colored bar corresponds to one dataset-specific adapter. The parameters of these domain-specific adapters are optimized for individual datasets. After the adaptation, the produced features can be considered the final universal representation and optimized for each dataset. To reduce the number of total parameters and model sizes, we design the whole network in a way that the basic feature extractors contain the majority of parameters, and the number of parameters in adapters is as few as possible. Formally, the whole network learning can be expressed as

$$\mathbf{W}_{\text{bs}}^*, \mathbf{W}_t^*\big|_{t=1}^T = \underset{\mathbf{W}_{\text{bs}}, \mathbf{W}_t\big|_{t=1}^T}{\arg\min} \frac{1}{T} \sum_{t=1}^{T} \frac{1}{N_t} \sum_{i=1}^{N_t} l\left(\mathsf{F}\left(x_i^t; \mathbf{W}_{\text{bs}}, \mathbf{W}_t\right), y_i^t\right) \tag{3}$$

where $\mathbf{W}_{\text{bs}}$ represents the parameters of the base network shared by all domains and $\mathbf{W}_t$ denotes the parameters in

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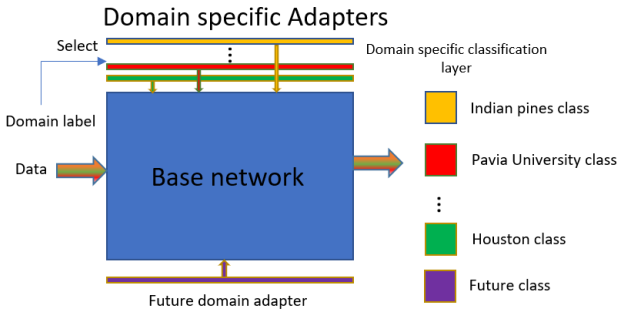


Fig. 2. Illustration of multidomain schemes.

the tth adapter. The number of parameters in $\mathbf{W}_{\text{bs}}$ should be significantly larger than that in $\mathbf{W}_t$. For the classification problem, two commonly used loss functions are hinge loss and cross-entropy loss. Since we are working in the deep learning context, cross-entropy loss with a softmax function is chosen for l. For the universal representation learning task, the model is trained on multiple hyperspectral datasets. It is possible that the number of training samples from different datasets varies significantly. In this case, the datasets with more samples might contribute more to the loss than the small datasets, and the training might be dominated by the large datasets. To address this issue, we feed the datasets to the model in a round-robin manner, as described in the experiment section, so that all datasets can receive sufficient training.

Fig. 2 further illustrates the multidomain classification framework. The blue rectangle represents the shared base network. The large size of the blue rectangle indicates that it accounts for the majority of the parameters. The domain-specific modules are demonstrated with different colors. The input data are sent to the classification network with multiple adapters. Each time only the adapter corresponding to the current data domain is selected and participates in the computation for adapting the base network. The domain label of the current data point can be easily obtained as prior knowledge or predicted by a classifier from the input data. An auxiliary network trained for domain classification can be used for this purpose. After obtaining the universal representations, the classification layer outputs the predicted class labels for the corresponding dataset at the end of the network. In the whole framework, the base network learns the shared knowledge among all domains, and the domain adapters are to capture domain-specific knowledge for adapting domain shifts with a small number of parameters. When we have multiple existing datasets at hand, the network can be trained on them simultaneously and perform multidomain classification.

To fully explore the representation ability of the universal network and optimize both shared base network and domain-specific adapters to individual datasets, we propose a two-stage training strategy to train the network. Specifically, in the first step, the whole universal network is trained on all datasets simultaneously with the optimization target achieving the highest average performance on them. The simultaneous training of multiple datasets can find the best solution for the parameters of the shared base network. However, it does not guarantee optimal domain-specific adapters because the optimization target in (2) is the average of all datasets. Thus, in step two, after the simultaneous training on all datasets, we freeze the parameters of the shared base network and further train the domain-specific part on each dataset individually. In addition, when a new dataset becomes available in the future, we only need to add and train a new adapter on top of the base network, as shown in Fig. 2, which is straightforward and does not require the retrain of the whole network.

The shared network can be ordinary CNN including 1-D, 2-D, and 3-D CNNs. There are two difficulties in the network design of hyperspectral multidomain classification. First, different HSI datasets usually have different numbers of classes. Even if two hyperspectral datasets have the same number of classes, very few of them share the same label spaces. This makes it impossible to share the final classification layer across datasets. To address this issue, we adopt a dataset-specific classification layer for our design, as shown in Fig. 2. In other words, each data domain has its own unique classification layer. Second, different hyperspectral datasets may have different spectral dimensions. Thus, the network should be able to deal with varied input sizes. For natural image multidomain classification, it is common to resize the images of different resolutions to make them compatible. Hyperspectral bands, however, have a clear physical meaning, so it is not appropriate to simply resize the spectral vector. To solve this problem, we employ the global average pooling layer after the last convolution layer so that the classifier can receive the same size input.

C. Domain Adapters for Multidomain HSI Classification

In this section, we present our domain adapter for learning universal representation in MHIC. The adapters have parameters specifically optimized for each domain and should be able to compensate for domain shift and learn domain-specific knowledge. First, we need to find out what should be adapted to address the distribution shift in multiple domains. In the training process of the same domain, the distribution at the input layer is normalized by preprocessing. However, as the training goes on, the weights of the network are changing. This varies the feature distribution of the hidden layers. From the perspective of a hidden layer, the distribution of its input is not the same, so they have to adapt to varying distributions [51]. This problem brings difficulty to network training and slows down the convergence. BN is a method proposed to alleviate the internal covariate shift in hidden layers of deep networks [51]. This method adjusts the mean and variance of hidden features to 0 and 1, and then learns a set of affine parameters to rescale and bias them per minibatch.

Motivated by the idea of BN, we propose to adapt our base network by domain-specific statistics and scaling factors. Mathematically, the computation of BN adaptation has four steps. First, we compute the domain-specific mean and variance for a minibatch with N_B samples

$$\mu_B^t = \frac{1}{N_B}\sum_{i=1}^{N_B} h_i^t \tag{4}$$

$$\sigma_B^{2^t} = \frac{1}{N_B}\sum_{i=1}^{N_B}\left(h_i^t - \mu_B^t\right)^2 \tag{5}$$

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where B represents a minibatch. t is the index of domains, so μ_B^t and $\sigma_B^{2^t}$ approximate domain-specific mean and variance. h is the hidden feature, and we ignore the index of feature position for simplicity. Then, we normalize the features per minibatch by the statistics of each domain

$$\hat{h}_i^t = \frac{h_i^t - \mu_B^t}{\sqrt{\sigma_B^{2^t} + \epsilon}} \tag{6}$$

where ϵ is a small constant. After this normalization for all domains, the representation power of the network is significantly reduced. To recover the original expressive capability, we add a scaling factor γ and biasing factor β operating on the normalized features for each domain

$$g_i^t = \gamma^t \hat{h}_i^t + \beta^t \tag{7}$$

where g_i^t represents the output of BN adapters. γ^t and β^t are trainable weights and learned along with other parameters in the network. Similar to mean and variance, the scaling and biasing factors are also domain-specific as indicated by the superscript t in the expression. Thus, the BN adaptation can handle different distributions of multiple domains.

We find in the experiment that adapting BN parameters with all spatial $3 \times 3 \times 3$ convolutions shared leads to significant performance gain compared to simply sharing a network for all domains. Works from image style transfer and domain adaptation show that the semantic information of images is modeled by the spatial convolution, while the style-related information (covariate shift) is mainly described by the mean and variance of feature maps in a deep CNN [52], [53], [54]. The proposed domain-specific affine parameters of BN address the style shift, thus, allow the spatial convolution to be shared, and focus on learning better style invariant semantics. The power of the BN adapter comes from its ability to compensate for the distribution shift of multiple domains. Assume that T hyperspectral data domains follow distribution $P_1(x_1, y_1)$, $P_2(x_2, y_2), \ldots, P_T(x_T, y_T)$, respectively. A single shared network (SSN) performs poorly on multiple datasets because of the distribution shifts. When we pass these data domains to a stack of convolution layers, the extracted feature maps follow different distributions as well. According to [55], the features of the hidden layer follow a Gaussian distribution. Therefore, for a certain hidden layer in the network, we obtain the distribution of neurons $N(\mu_1, \sigma_1^2), N(\mu_2, \sigma_2^2), \ldots, N(\mu_T, \sigma_T^2)$ for domains $1, 2, \ldots, T$. If we use domain-shared BN, the mean and variance will be calculated on all datasets, producing μ_{shared} and σ_{shared}^2. Furthermore, the features with originally different distributions will be rescaled and biased by the same affine parameter γ_{shared} and β_{shared}, distorting the distributions by forcing different domains to have the same mean and standard deviation. It is obvious that this will harm the model's learning ability. In the proposed BN adapter, we allow each domain to have its specific γ^t and β^t. This gives the network an opportunity to recover domain-specific distributions.

The BN adaptation can adapt the distribution of features in BN layers. However, it only has two learned parameters, i.e., γ and β. Such a small number of parameters of the adaptive module might not be powerful enough to capture all domain-specific knowledge for each dataset. Furthermore, it may not completely mitigate the large domain shift in different hyperspectral datasets. Since most important features are extracted by convolution layers, it is desirable to impose adaptation on convolution layers. To enhance the adaptability of the MDN, we insert additional parameters to further adapt the base network. Specifically, these adapting components are placed in parallel with normal convolution layers and added to them. They are a bank of domain-specific convolution adapters. To maximally reduce the number of parameters in these adapters, we adopt pointwise (PW) convolution as the domain-specific adapters. Formally, take 3-D convolution as an example; a basic computation module can be expressed as

$$g = k_{P\times P\times P} * h + k_{1\times 1\times 1}^t * h \tag{8}$$

where $*$ represents convolution, and $k_{P\times P\times P}$ and $k_{1\times 1\times 1}^t$ denote $P \times P \times P$ and $1 \times 1 \times 1$ convolution kernel banks, respectively. $k_{P\times P\times P}$ is shared by all domains, and $k_{1\times 1\times 1}^t$ is the domain-specific operation for the tth domain. The ratio of the number of parameters in this adapter to that in the base network is $1/(P \times P \times P)$. We can see that the adapter only introduces a small number of parameters. Here, we use the notations for 3-D convolution networks for illustration. Corresponding PW convolution for 1-D and 2-D convolution can be used in 1-D CNN and 2-D CNN as well. The PW convolution is equal to the linear combination of each feature channel, and the coefficients for the linear combination are the learned convolution kernels

$$m_s = \sum_{c=1}^{C} w_{\text{cs}}^t f_c \tag{9}$$

where m_s is the sth channel of the output, C is the number of input channels, w_{cs}^t is the kernel weights, and f_c is the cth input feature map. The convolution layers in the base network extract the common set of features for all datasets in different channels. However, different domains may require different subsets of features from all the base channels. The feature channels can contribute unequally to each data domain. Different linear combinations of them with different weights (combination kernels) for each domain explore more diversity and capacity of the domain-specific representation ability with very few parameters. In other words, different datasets have different cross-channel statistics, which needs to be compensated. Thus, the domain-specific PW convolution models the channelwise feature relationship for the individual dataset. Adapted with the domain-specific channelwise relationship, the final features have different distributions according to each dataset, so the universal network is able to better capture domain-specific knowledge for individual datasets.

We can impose independent weight decay and other regularizations on the domain-specific PW convolution. In other words, different strengths of regularization can be added to different domain adapters. This provides an opportunity to tune the coefficient of regularization according to the training data sizes of each domain. When training datasets are small, stronger regularization can be executed on the adapting module to avoid overfitting. If the weight decay coefficient is very high, the PW convolution adapters can reduce to zero.

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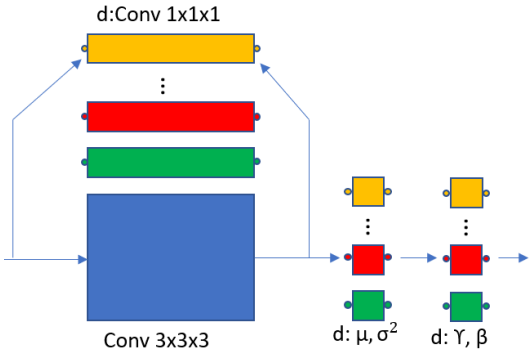


Fig. 3. Illustration of a convolution block with adapters.

Fig. 3 shows a convolution block with the proposed adapting mechanisms. The blue rectangle represents convolutional computation shared across all domains. The $1 \times 1 \times 1$ convolution, batch mean and variance, and affine parameters are domain-specific as denoted by d : in front.

With the proposed adaptive module, our model can be trained on multiple datasets without conflicts and learns powerful universal representations. An advantage of this paradigm is that it increases the number of samples that the model sees. Although different datasets have their own characteristics, joint training may bring benefits when several datasets have common properties. In hyperspectral remote sensing, some datasets have fewer labeled samples than others. In this context, the smaller datasets can benefit from larger datasets when they have common properties. At the same time, the shared network parts learn from more diversified data with different scales and spectral properties. This helps the model to learn better in-variances.

In addition, when a new dataset is collected in the future, we only need to add one more PW convolution and BN adapter on top of the base network and train the new parameters with the base network frozen. This not only fully utilized abundant knowledge learned from all previous datasets in the base network but also helped prevent overfitting on the new datasets. Since the newly introduced parameters occupy a tiny proportion of the base networks, the number of parameters of the universal network does not increase much. It is particularly useful when the new domain has a limited number of training samples.

IV. Experiments and Analysis

In this section, we present experiments conducted to evaluate the proposed universal representation learning network for MHIC. First, we create an MHIC benchmark by combining seven public available hyperspectral datasets. Then, we describe the details of experiments, including configurations and compared approaches. The results are shown and analyzed to prove the effectiveness of our method.

A. Benchmark Datasets for Hyperspectral Multidomain Classification

While some multidomain learning benchmarks are available in computer vision and medical imaging, to the best of our knowledge, there is no such benchmark for HSI classification. Therefore, we create a benchmark called MHIC dataset.

The MHIC benchmark was created by assembling seven existing hyperspectral remote sensing datasets. These include Indian Pines, Salinas, Pavia University (PaviaU), Kennedy Space Center (KSC), Botswana, Pavia Center (PaviaC), and University of Houston (UHouston) datasets. These datasets cover a wide diversity of hyperspectral data domains with different sensor types, the number of bands, spatial resolutions, land cover types, and image classes.

Indian Pines was collected by AVIRIS sensor with 224 bands. In the publicly available dataset, 24 noisy and water absorption bands were removed with 200 bands left. The original dataset has 16 land cover categories. The number of labeled samples per class is severely unbalanced. Some have a very limited number of samples. This brings a big challenge for machine learning models. For example, the class Oats and Grass-pasture-mowed have only 20 and 28 classes, respectively, while Soybean-mintill class has 2455, which is more than 100 times the class Oats. Salinas was also acquired by the AVIRIS sensor. It has 16 classes, each with abundant labeled samples. The dataset provider removed 20 useless bands, leaving 204 for analysis. KSC is another widely used dataset captured by the AVIRIS sensor. It has 13 labeled classes and 176 bands after noisy band removal.

PaviaU and PaviaC datasets are high-resolution hyperspectral datasets with 1.3-m spatial resolution. The ROSIS sensor was used to scan an urban area in Italy. There are 103 bands and 102 bands for the PaviaU and PaviaC datasets, respectively. The Botswana dataset was collected by a satellite project EO-1. It has a lower spatial resolution of 30 m. 145 bands were used for classification. The Botswana dataset has 14 classes. The UHouston dataset was released for the IEEE GRSS data fusion contest in 2013, containing both hyperspectral and LiDAR data. We use the hyperspectral part only. It was acquired in the UHouston campus and surrounding areas with 15 classes manually labeled by a human. This hyperspectral data have 144 bands ranging from 380 to 1050 nm. The flying height of the acquisition was 5500 ft, and the spatial resolution of the UHouston dataset is 2.5 m.

Tables I and II show the key parameters, the category names, and the number of samples per category for all seven hyperspectral datasets adopted in the MHIC benchmark. The benchmark covers as much data domain diversity as possible. It includes three different remote sensing platforms and sensors: AVIRIS, ROSIS, and EO-1. For the same sensor of AVIRIS, different numbers of bands are presented: 200, 204, and 176 for Indian Pines, Salinas, and KSC datasets, respectively. Different spatial resolutions also exist in these datasets: 1.3 m for PaviaU, 3.7 m for Salinas, 18 m for KSC, 20 m for Indian Pines, and 30 m for Botswana. Indian Pines and Salinas datasets contain many vegetation land cover types, while PaviaU and PaviaC contain more urban structures. KSC and Botswana have mixed and complex geological types. Comparing the total number of labeled samples, KSC and Botswana datasets have significantly less number of samples than the other five datasets. With such big domain diversity in these seven datasets, it is very challenging to design a single

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TABLE I
KEY PARAMETERS OF SEVEN DATASETS

	Indian Pines [56]	Salinas [57]	PaviaU [58]	KSC [57]	Botswana [57]	PaviaC [58]	UHouston [59]
Spectral range (nm)	400-2500	400-2500	430-960	400-2500	400-2500	430-960	380-1050
No. of bands used	200	204	103	176	145	102	144
Spatial resolution (m)	20	3.7	1.3	18	30	1.3	2.5
Sensor	AVIRIS	AVIRIS	ROSIS	AVIRIS	EO-1	ROSIS	NCALM

TABLE II
CLASSES AND NUMBER OF SAMPLES OF SEVEN DATASETS USED FOR EXPERIMENTS

Indian Pines		**Salinas**		**PaviaU**		**KSC**		**Botswana**		**PaviaC**		**UHouston**	
Class	**No.**	**Class**	**No.**	**Class**	**No.**	**Class**	**No.**	**Class**	**No.**	**Class**	**No.**	**Class**	**No.**
Alfalfa	46	Brocoli-1	2009	Asphalt	6631	Cabbage-p	256	Water	270	Water	65971	Healthy-g	1251
Corn-n	1428	Brocoli-2	3726	Meadows	18649	oak-h	252	Hippo-g	101	Trees	7598	Stressed-g	1254
Corn-m	830	Fallow	1976	Gravel	2099	Cattail-m	404	Floodplain-1	251	Asphalt	3090	Synthetic-g	697
Corn	237	Fallow-r	1394	Trees	3064	Graminoid-m	431	Floodplain-2	215	Bricks	2685	Trees	1244
Grass-p	483	Fallow-s	2678	metal-s	1345	Hardwood-s	105	Reeds1	269	Bitumen	6584	Soil	1242
Grass-t	730	Stubble	3959	Soil	5029	Mud-f	503	Riparian	269	Tiles	9248	Water	325
Grass-m	28	Celery	3579	Bitumen	1330	broadleaf-h	229	Firescar2	259	Shadows	7287	Residential	1268
Hay-w	478	Grapes-u	11271	Bricks	3682	Salt-m	419	Island-i	203	Meadows	42826	Commercial	1244
Oats	20	Soil-v	6203	Shadows	947	Scrub	761	Acacia-w	314	Soil	2863	Road	1252
Soybean-n	972	Corn-w	3278			Slash-p	161	Acacia-s	248			Highway	1227
Soybean-m	2455	Lettuce-4	1068			Spartina-m	520	Acacia-g	305			Railway	1235
Soybean-c	593	Lettuce-5	1927			Water	927	Short-m	181			Parking-1	1233
Wheat	205	Lettuce-6	916			Willow-s	243	Mixed-m	268			Parking-2	469
Woods	1265	Lettuce-7	1070					Exposed-s	95			Tennis-c	428
Buildings	386	Vinyard-u	7268									Running-t	660
Stone	93	Vinyard-v	1807										
Total	10249	**Total**	54129	**Total**	42776	**Total**	5211	**Total**	3248	**Total**	148152	**Total**	15029

model that can simultaneously solve classification problems on all of these datasets.

B. Multidomain Classification of 3-D CNN

We first conduct experiments of multidomain hyperspectral classification using spectral–spatial CNNs. Since our goal is to fit multiple challenging hyperspectral datasets using one single model, the chosen backbone network should have sufficient capacity for such a task. In both computer vision and HSI classification, most state-of-the-art networks choose ResNet [60] as their base backbone [61]. One reason for this is that ResNet has a large capacity for many complex tasks. By eliminating fully connected layers, which have a majority of parameters, ResNet can take full advantage of all their parameters and reduce the risk of overfitting. For these reasons, we choose 3-D ResNet as our backbone network architecture for the experiments.

Our network architecture follows the setting in [13] but replaces their 2-D convolution with 3-D convolution since we need spectral–spatial feature extraction. In general, our ResNet has three stages stacked together. The spectral dimension of feature maps in each stage is reduced half by downsampling, while the spatial dimension remains the same until the global average pooling layer. We use convolution with a stride of 2 in the spectral dimension and 1 in the spatial dimension at the first convolution of each stage to achieve it. Feature maps within one stage have the same size. The numbers of feature maps for each stage are 64, 128, and 256, respectively. Before stage 1, we use a first convolution layer with 32 feature maps. In each stage, we use four residual blocks of basic residual units. Each residual unit consists of two convolution layers and a shortcut connection. All convolutions in the backbone network use a kernel of size $3 \times 3 \times 3$. The convolutional block is implemented as in Fig. 3. Specifically, the output features from the previous layer go through a shared convolution layer with $3 \times 3 \times 3$ convolution kernel. At the same time, the same features are input into seven parallel convolution layers of $1 \times 1 \times 1$ kernel, each parallel branch for one data domain. Then, the parallel branches are merged by an addition operation. Finally, the merged feature maps are sent through another seven parallel operations of BN. One BN corresponds to one data domain. The selection of parallel branches according to the input data domain is implemented by Pytorch's ModuleList. At the end of all convolution layers, 3-D global average pooling is used to obtain the final universal feature vector. The last layer is the domain-specific classification layer with cross-entropy loss.

To evaluate the effectiveness of our proposed network in the task of multidomain classification, some conventional paradigms are chosen as the competitors. The first baseline is multiple individual networks (MINs). This is the traditional method that trains seven independent classification networks from scratch on the seven datasets. This leads to a significant cost with seven independent models. The second comparison method is an SSN with all convolution layers completely shared by all seven domains. This scheme has the smallest total model size. Note that, in this case, there are still domain-specific layers as classification layers due to different numbers of categories for different datasets.

We also present the experimental results of our proposed MDN with adapters and analyze the number of parameters to be added when introducing one more domain. Suppose that the number of channels of the rth convolution layer is C_r and the convolution kernel size is $P \times P \times P$. The number of parameters in a normal convolution layer is $P \times P \times P \times C_{r-1} \times C_r$ (ignoring bias terms). The increased number of parameters in the BN layer of adapters is $2 \times C_r$. Usually,

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TABLE III
COMPARISON OF OA, AA, AND KAPPA COEFFICIENT ON 3-D CNNS WITH EXPERIMENT SCHEME 1

	OA(%)					AA(%)					Kappa(X100)				
	MIN	SSN	RA	ASPIN	MDN	MIN	SSN	RA	ASPIN	MDN	MIN	SSN	RA	ASPIN	MDN
IP	98.37	35.87	98.06	93.73	**98.82**	84.70	19.91	94.43	71.10	**97.90**	98.13	24.21	97.79	92.83	**98.65**
Sal	**99.97**	10.69	99.72	97.02	99.83	**99.99**	11.50	99.88	97.65	99.90	**99.97**	2.97	99.69	96.68	99.81
PU	**99.95**	98.64	99.79	98.31	99.92	98.89	97.73	99.64	96.58	**99.83**	**99.94**	98.18	99.72	97.76	99.89
KSC	99.30	50.63	98.94	96.93	**99.40**	99.13	51.17	98.42	95.11	**99.17**	99.22	45.62	98.82	96.58	**99.33**
Bot	94.38	39.67	96.96	98.50	**98.92**	94.66	41.38	97.15	98.21	**98.66**	93.92	35.09	96.71	98.37	**98.83**
PC	**99.98**	87.61	99.87	98.18	99.86	**99.96**	61.62	99.64	92.31	99.57	**99.98**	81.71	99.81	97.42	99.80
UH	98.76	68.14	99.13	95.73	**99.03**	98.51	64.00	99.03	94.73	**98.90**	98.66	65.47	**99.06**	95.39	98.95
Avg	98.67	55.89	98.92	96.91	**99.40**	96.55	49.62	98.31	92.24	**99.13**	98.55	50.46	98.80	96.43	**99.32**

this amount of increase is less than 1% of the total number of parameters. The other part of the adapter that introduces new parameters is the domain-specific PW convolution. The number of parameters in the domain-specific PW convolution is $C_{r-1} \times C_r$. Thus, the percentage of added parameters per domain for the PW convolution is $1/P \times P \times P$. When $P = 3$, the ratio is about 3.7%. To compare the MDN with other multidomain learning methods, we evaluated the performance of RA [41] and ASPIN [49] on the hyperspectral datasets. RA proposes a domain adapter in a serial residual manner, and ASPIN uses instance normalization rather than BN for adaptation.

The only preprocessing step required for the seven datasets is to normalize each dataset to a mean of 0 and a standard deviation of 1. To train the models and test their performance, we randomly split each dataset into a training set and a testing set. The individual datasets are split according to each category. We conducted experiments on two different split and dataset sampling schemes. For general comparison, in scheme 1, we split 20% of each dataset for training, and the rest are used for testing purposes. The patch size of a sample is chosen as 19. For evaluation metrics, we use the overall accuracy (OA) of all classified samples, average accuracy (AA) per class, and Kappa coefficient of the results on the test sets to evaluate the individual dataset. For the total evaluation of all data domains, the metrics averaged over the seven datasets are adopted.

One challenge of training a multidomain model on several datasets simultaneously is to balance different datasets in the training process. Most neural network training schemes are batch-based. Each iterative step is performed based on the gradient of a minibatch of samples. When optimizing multiple data distributions, we still create the batches from a single pure data domain. The reason is that domain-specific BN layers are used in our network, which requires a pure batch of the domain to calculate the mean and variance for BN. Moreover, different datasets have different numbers of training samples. Thus, datasets with a larger number of training samples might have more influence on the model if all datasets are treated equally. In our experiment, the number of samples in the Salinas dataset is more than ten times that in the Botswana dataset. To make each dataset contributes equally to the training process, the data are fed to the network in a round-robin manner. Specifically, a pure batch of training data from each dataset is fed to the network in turn. This operation has the same effect as assigning higher weights to smaller datasets for the purpose of balancing. We assume that the domain label of a batch of data is known. When this prior information is not available, an auxiliary network can be trained to predict the domain label.

We used stochastic gradient descent with momentum to train all the methods in our experiment. The initial learning rate is 0.1 and reduced by ten times on the plateau in the training process. The momentum is set as 0.9. The total number of training epochs is 1000. We also use weight decay and dropout on the network to control the capacity of our models. During training, we used the following types of online data augmentation: random horizontal flip or vertical flip with 50% probability, adding simulated random radiation noise, and adding simulated mixture and noise. A similar training strategy is used to train the competitor algorithms.

Table III demonstrates the OA, AA, and Kappa coefficient of the proposed MDN and the competitors on the seven datasets. We can see that MIN is a very strong baseline with 98.67% average OA, 96.55% average AA, and 98.55% average Kappa across all datasets. MIN has the largest model size with seven times the number of parameters of an individual model. SSN is the naive way of sharing parameters for all datasets in one model and has only the same number of parameters of an individual model. However, without any domain adapter mechanisms, it performs poorly because of domain shifts among different datasets. This proves the significance of the proposed domain adapters to mitigate domain shifts. A minor adapting of features toward different distributions has boosted performance significantly. The proposed MDN outperforms SSN by a large margin in all three averaged metrics and achieves an average performance of 99.40% OA, 99.13% AA, and 99.32 Kappa coefficient. RA and ASPIN also significantly outperform SSNs because of their introduced domain adapters. This again shows the importance of the adaptation mechanism for multidomain learning. However, their performances are still lower than MDN. This proves that the design of the adapter has a high influence on multidomain learning.

We further find that the average OA, AA, and Kappa of MDN are also better than the seven single-model banks. A possible reason for this improvement is that the MDN sees more data, and the seven datasets contain some shared knowledge that can be utilized by our shared multidomain model. In addition, by training on seven HSI datasets of different scales and radiation conditions, the multidomain model has the potential to learn more invariant features. On the seven datasets, the MDN has less than 1.3 times the number of parameters of an individual network (feature extraction part

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TABLE IV
COMPARISON OF OA, AA, AND KAPPA ON 3-D CNN WITH EXPERIMENT SCHEME 2

	OA(%)					AA(%)					Kappa(×100)				
	MIN	SSN	RA	ASPIN	MDN	MIN	SSN	RA	ASPIN	MDN	MIN	SSN	RA	ASPIN	MDN
IP	79.27	20.80	81.08	80.55	**83.33**	87.43	16.63	88.53	77.94	**91.59**	76.39	11.61	78.41	77.74	**80.92**
Sal	85.88	23.78	87.59	89.89	**91.42**	93.27	26.82	93.53	94.23	**95.49**	84.38	19.44	86.21	88.74	**90.45**
PU	82.54	66.43	87.32	87.48	**89.95**	88.29	77.11	91.50	91.52	**93.26**	77.42	58.95	83.50	83.64	**86.85**
KSC	94.06	31.18	93.86	88.99	**95.17**	91.43	26.73	90.97	81.85	**92.15**	93.32	24.77	93.10	87.61	**94.57**
Bot	95.01	21.26	**97.84**	96.22	97.43	96.10	22.32	**98.39**	95.89	98.01	94.57	13.88	**97.64**	95.88	97.20
PC	96.72	91.88	**98.08**	97.20	97.54	94.69	76.50	**96.79**	94.60	95.82	95.37	88.43	**97.29**	96.03	96.52
UH	88.69	16.32	89.41	84.73	**92.53**	89.80	17.09	90.62	86.86	**93.28**	87.75	9.16	88.53	83.46	**91.91**
Avg	88.88	38.81	90.74	89.29	**92.48**	91.57	37.60	92.90	88.98	**94.23**	87.03	32.32	89.24	87.59	**91.20**

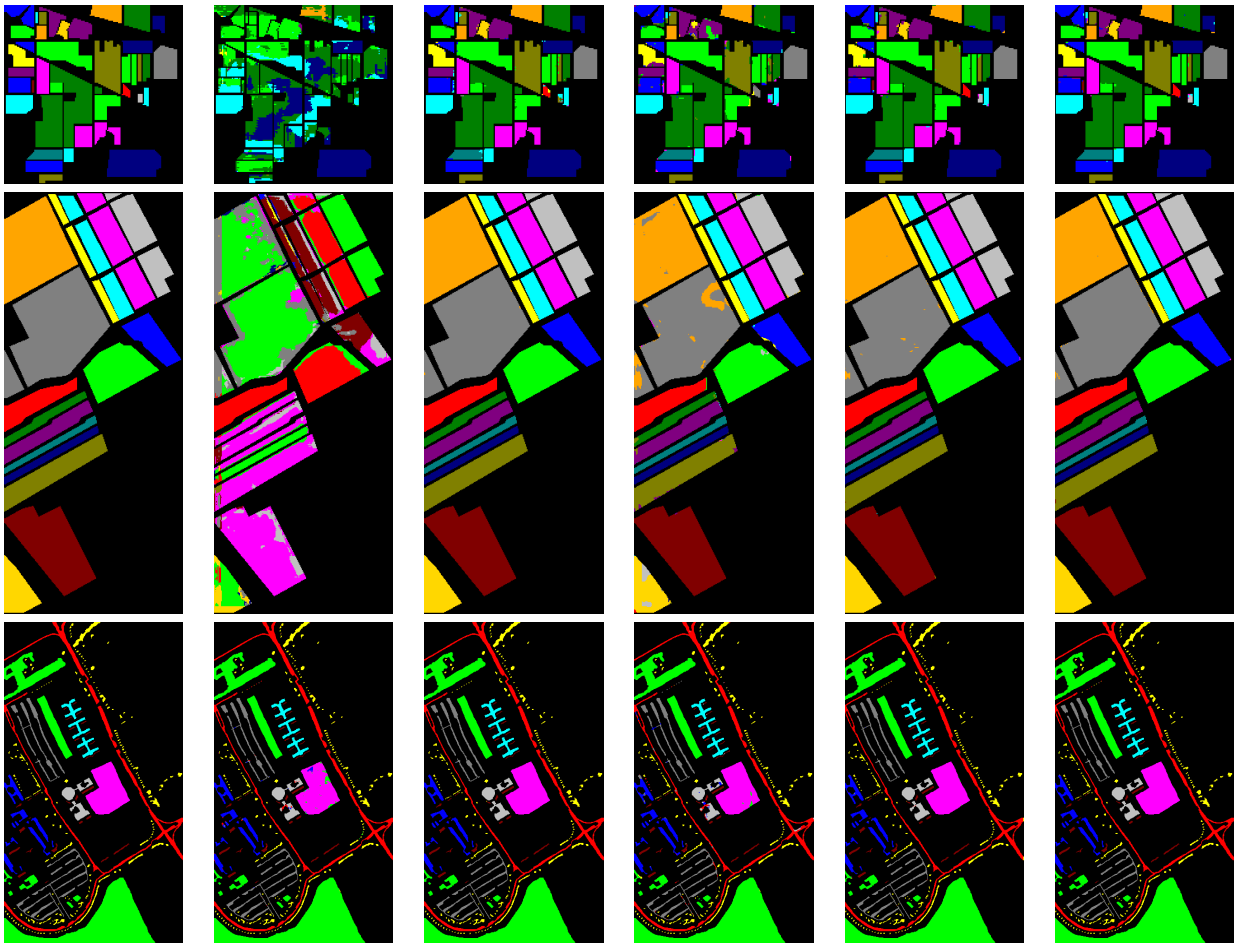


Fig. 4. Ground truth and classification maps of 3-D CNN with experiment scheme 1. From (top) to (bottom) Indian Pines, Salinas, and PaviaU. From (left) to (right) ground-truth classification map and results from SSN, MIN, ASPIN, RA, and MDN, respectively.

of the network before the classification layer). Comparing the performance of individual datasets, we can find that MDN performs on par with or better than MIN. On the Botswana dataset, MDN gains a particularly large performance increase over MIN. Classification maps of this experiment are shown in Fig. 4.

To further prove the effectiveness of the proposed MDN with a smaller patch size and fewer training samples, we conducted the second comparison study in sampling scheme 2. Specifically, the patch size of a sample is chosen as five, so a smaller area of spatial features is used by the 3-D CNN. For the Salinas, PaviaU, PaviaC, and UHouston datasets, we randomly sample 50 samples per class as the training set and 50 samples per class as the validation set. For KSC and Botswana datasets that have fewer samples, we randomly sample 30 samples per class as the training set and 30 per class as the validation set, respectively. In the Indian Pines dataset, some classes have even fewer samples so that it is infeasible to sample 30 from them. For such classes, smaller numbers were sampled as training and validation sets: 15 for each training and validation for Alfalfa, 10 for Grass-m, and 7 for Oats. Stone has sufficient samples, so we sampled 30 for Stone. For the rest of the classes in the Indian Pines dataset, the sampling scheme was the same as the Salinas dataset. All samples in the datasets not included in the training and validation sets were used as the test set.

Table IV shows the OA, AA, and Kappa of the classification results from multiple individual models, a single shared model, and the proposed multidomain model. We can see that all evaluation metrics are lower than their counterparts

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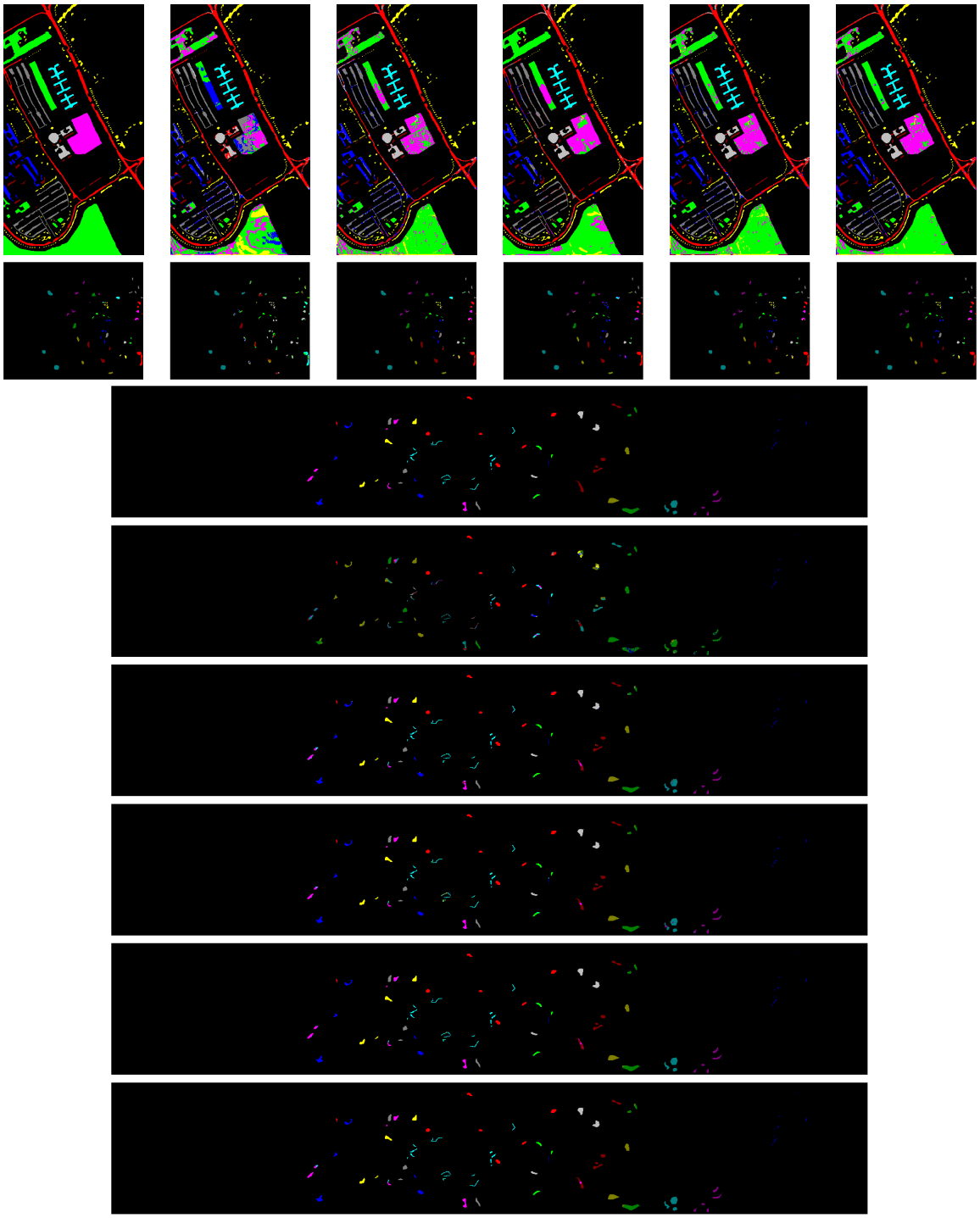


Fig. 5. Ground truth and classification maps of 3-D CNN with experiment scheme 2. The first two rows: PaviaU and KSC datasets. From (left) to (right) ground-truth classification map and results from SSN, MIN, ASPIN, RA, and MDN, respectively. The third to eighth rows: ground-truth classification map on the Botswana dataset and results from SSN, MIN, ASPIN, RA, and MDN, respectively.

in experiment scheme 1. There are two reasons for this: 1) in experiment scheme 2, a smaller spatial context can be used for classification and 2) fewer training samples are used to train the models. Comparing MIN, SSN, and MDN, we can see the same trends that MDN outperforms the other two methods in average performance on the seven datasets. Specifically, MDN outperforms MIN by 3.60% in average OA, 2.66% in average AA, and 4.17 in average Kappa. For performance on individual datasets, we can find that MDN performs the best on all seven individual datasets. This proves the effectiveness of the proposed MDN in fewer training samples and less spatial context. RA and ASPIN

TABLE V

COMPARISON ON THE NUMBER OF PARAMETERS FOR SEVEN DATA DOMAINS

	MIN	SSN	RA	ASPIN	MDN
#parameters	7×	1×	1.26×	1.01×	1.26×

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TABLE VI
COMPARISON OF OA, AA, AND KAPPA ON 1-D CNN

	OA (%)				AA (%)				Kappa (×100)			
	MIN	SSN	NPW	MDN	MIN	SSN	NPW	MDN	MIN	SSN	NPW	MDN
Indian Pines	65.31	12.34	70.76	**73.03**	74.72	14.97	**77.94**	74.17	60.78	3.3	66.80	**68.99**
Salinas	87.70	13.77	**88.10**	87.94	93.74	19.70	93.75	**94.22**	86.32	10.59	**86.74**	86.57
Pavia University	83.43	52.30	84.20	**87.76**	86.83	73.58	87.05	**89.98**	78.34	44.30	79.30	**83.99**
KSC	89.53	11.53	89.82	**92.21**	85.05	14.15	85.18	**89.07**	88.24	5.38	88.56	**91.25**
Botswana	92.15	17.23	91.81	**93.02**	93.45	20.13	93.32	**94.31**	91.46	11.44	91.10	**92.41**
Pavia Center	96.05	87.79	96.45	**97.74**	92.17	75.27	92.30	**94.85**	94.42	83.10	94.97	**96.80**
University of Houston	86.98	12.61	83.79	**89.05**	86.74	12.11	83.82	**88.59**	85.90	5.76	82.45	**88.13**
Average	85.88	29.65	86.42	**88.68**	87.53	32.84	87.62	**89.31**	83.64	23.41	84.27	**86.88**

TABLE VII
COMPARISON OF MDN, RA, AND ASPIN ON 1-D CNN

	OA(%)			AA(%)			Kappa(×100)		
	RA	ASPIN	MDN	RA	ASPIN	MDN	RA	ASPIN	MDN
Indian Pines	71.14	67.32	**73.03**	**79.09**	76.40	74.17	67.23	62.92	**68.99**
Salinas	86.52	86.28	**87.94**	93.19	93.24	**94.22**	85.01	84.76	**86.57**
Pavia University	80.33	84.32	**87.76**	85.56	87.98	**89.98**	74.45	79.66	**83.99**
KSC	89.89	90.16	**92.21**	85.10	86.07	**89.07**	88.64	88.95	**91.25**
Botswana	92.19	91.32	**93.02**	93.56	92.95	**94.31**	91.51	90.56	**92.41**
Pavia Center	96.02	96.08	**97.74**	92.20	92.19	**94.85**	94.38	94.46	**96.80**
University of Houston	84.63	85.19	**89.05**	84.62	85.30	**88.59**	83.35	83.98	**88.13**
Average	85.82	85.81	**88.68**	87.62	87.73	**89.31**	83.51	83.61	**86.88**

consistently perform better than MIN on average OA and Kappa. Our MDN still demonstrates an advantage with at least 1.74% OA, 1.33% AA, and 1.96 Kappa higher than the two multidomain methods on the seven datasets. The corresponding classification maps are shown in Fig. 5.

Following Rebuffi et al. [41], we analyze the number of parameters of MDN, RA, ASPIN, SSN, and MIN (excluding the classification layer because they are impossible to be shared) and list the results in Table V to compare the spatial efficiency. The base unit is the number of parameters of SSN. As shown in Table V, MIN is seven times the size of SSN. The most efficient methods are SSN and ASPIN. MDN has significantly fewer parameters than MIN with only 1.26 times the parameters of SSN, the same as RA.

To compare the time efficiency, we evaluate the inference time of a single testing data sample for the comparison models on a Tesla V100 GPU. The results show that MDN and a single network have nearly the same inference time. The reason is that the domain-specific module only introduces less than 3.7% (for seven datasets, it accumulates to 25.9%) more parameters than a single network for one dataset, and the operations are a very lightweight linear combination of feature maps.

C. Multidomain Classification of 1-D CNN

In this section, we evaluate the performance of the proposed MDN for spectral classification. This is a challenging scenario because no spatial context is used for extracting features. It is a more practical situation in remote sensing applications and avoids the potential overlap of training patches and testing patches. To perform spectral classification, we adapted the CNNs and the multidomain design to its 1-D version. Specifically, a similar ResNet architecture is adopted with all 3-D convolution replaced with 1-D convolution. All convolution kernel sizes are chosen as 3. BN and PW convolution operations are conducted in a 1-D manner. In this experiment, we also conducted an ablation study when only BN layer is adapted. In this case, the number of parameters of the MDN is nearly the same as an SSN. The dataset split strategy follows scheme 2.

Table VI shows the results of different solutions for multidomain classification. The NPW represents the MDN with no PW convolution adapter (adapting BN only). The results demonstrate that the domain-specific BN adapter is very powerful at multidomain classification. NPW increases very few parameters to the SSN but significantly outperforms the latter by 56.77% in average OA, 54.78% in average AA, and 60.86 in average Kappa coefficient. This proves the utter importance of adapting domain shifts in multidomain classification. NPW also performs slightly better than MIN, which has nearly seven times the parameters of NPW. Combining the BN adapter and the PW convolution adapter, the proposed MDN achieves the best performance, beating MIN with 2.80% in average OA, 1.78% in average AA, and 3.24 in average Kappa coefficient with a smaller number of parameters. MDN also achieves the best performance in the majority of the individual datasets. Table VII further compares MDN with RA and ASPIN. MDN outperforms two competitors with a large margin on almost all datasets in the three metrics except the AA on Indian Pines. This clearly validates the superiority of the proposed MDN over RA and ASPIN. Fig. 6 shows the classification maps.

Research in the interpretation of deep learning shows that shallow layers learn low-level local features and deeper layers extract high-level semantic features [62], [63]. We further conducted experiments to analyze which layers in the network need to be adapted and explored the possibility of completely sharing some layers. Specifically, there are three stages in our ResNet architecture named beginning stage, middle stage, and last stage. Each time, we attach adapters on one stage and leave the other stages unadapted. This gives us three ablated versions of the MDN: adapt beginning, adapt middle, and adapt last.

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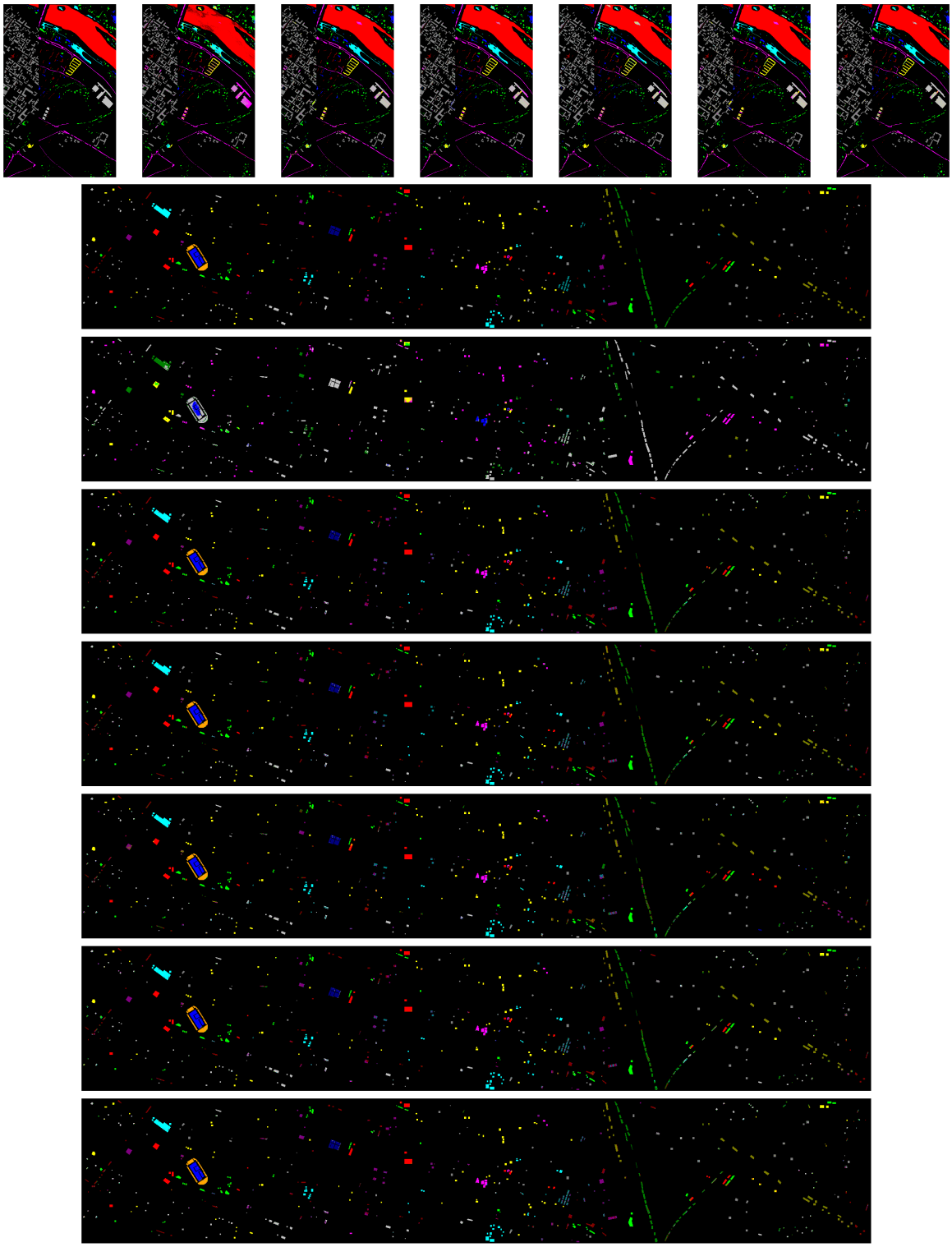


Fig. 6. Ground truth and classification maps of 1-D CNN. The first row, from (left) to (right), ground-truth map of the PaviaC dataset and classification maps of SSN, MIN, NPW, ASPIN, RA, and MDN, respectively. The second to eighth rows: ground-truth map of the Houston dataset and classification maps of SSN, MIN, NPW, ASPIN, RA, and MDN, respectively.

We compare these versions with fully adapted MDN. All other experiment setups follow the previous experiment.

The classification performance of each version and the full MDN is listed in Table VIII. We can see that adapting to only one stage causes a significant performance drop compared to adapting to all stages. All average metrics of OA, AA, and Kappa decrease by about more than ten points in the three ablated versions. Although adapting to the beginning stage leads to an average classification performance better than adapting to the middle stage and the last stage, the overall performance of adapting to the beginning is still poor. This shows the importance of adapting all stages for the good performance of multidomain learning. The performance on individual datasets demonstrates different trends. For example,

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TABLE VIII
ADAPT BEGINNING, MIDDLE, LAST, AND ALL LAST STAGES

	OA (%)				AA (%)				Kappa (×100)			
	Beginning	Middle	Last	All	Beginning	Middle	Last	All	Beginning	Middle	Last	All
Indian Pines	54.45	50.41	44.91	**73.03**	65.37	62.93	54.53	**74.17**	49.48	45.29	39.21	**68.99**
Salinas	82.56	85.27	75.47	**87.94**	86.19	92.30	82.30	**94.22**	80.56	83.66	72.82	**86.57**
Pavia University	74.48	79.80	82.19	**87.76**	78.81	81.79	85.77	**89.98**	66.61	72.91	76.70	**83.99**
KSC	73.71	42.41	44.55	**92.21**	69.44	33.75	40.79	**89.07**	70.58	37.09	39.04	**91.25**
Botswana	82.85	80.94	83.18	**93.02**	86.13	83.09	85.17	**94.31**	81.36	79.27	81.72	**92.41**
Pavia Center	95.37	96.28	95.94	**97.74**	90.94	91.28	91.20	**94.85**	93.45	94.73	94.27	**96.80**
University of Houston	72.47	70.81	71.99	**89.05**	74.11	72.93	73.55	**88.59**	70.24	68.48	69.71	**88.13**
Average	76.56	72.27	71.18	**88.68**	78.71	74.01	73.33	**89.31**	73.18	68.78	67.64	**86.88**

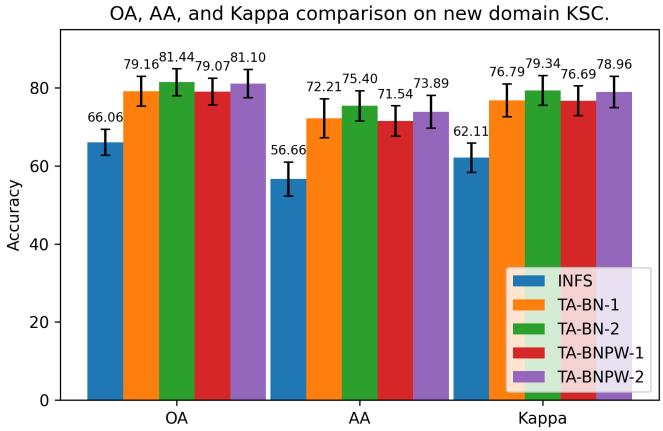


Fig. 7. OA, AA, and Kappa comparison on new domain KSC.

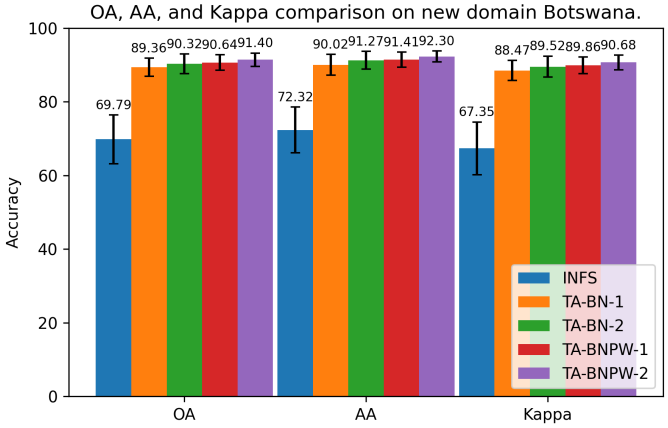


Fig. 8. OA, AA, and Kappa comparison on new domain Botswana.

adapting to the beginning stage is especially important for the KSC dataset. Without adapting to the beginning stage, the classification accuracy drops significantly to lower than 50%. The OA, AA, and Kappa on the PaviaC dataset experience the smallest drop when adapting only one stage.

D. MDN on New Small Domain

In Subsection IV-B and IV-C, the proposed MDN has been evaluated in the scenario of training and testing on multiple domains simultaneously. In this section, we consider another situation: when very few data from new domains become available, how well does an existing MDN generalize to the new domains. The same 3-D ResNet architecture with the proposed adapters in Section IV-C is used to conduct this experiment. We assume the five larger datasets: Indian Pines, Salinas, PaviaU, PaviaC, and UHouston as already available domains to train an MDN; KSC and Botswana datasets are used as the newly collected small domains. The reason for this choice is that the original KSC and Botswana datasets have much fewer samples than the other five datasets. Specifically, we consider a challenging case where only three training samples per class are available for KSC and Boswana for "three-shots" learning. First, we trained an MDN with five adapters on the five larger datasets and froze the parameters. Then, two randomly initialized new adapters for KSC and Botswana were attached on top of the existing trained base network. Only the two new adapters were trained on the two small datasets. We consider two cases of base models trained using 20% and 80% samples of the five datasets, respectively, and call them base1 and base2. For each base network and added adapters, we consider two versions: 1) training BN only and 2) training both BN and PW convolution. The resulting trained models on top of base1 are called training adapter BN-1 (TA-BN-1) and training adapter BNPW-1 (TA-BNPW-1). The models trained from base2 follow the same naming rule and are called TA-BN-2 and TA-BNPW-2. We also trained independent networks for the two small datasets from scratch (similar to MIN) called independent network from scratch (INFS) and compared the performance with the models from training adapters. We randomly split the datasets ten times and ran an experiment for each split. The mean and standard deviation of the evaluation metrics are reported.

Figs. 7 and 8 show the classification results of the competing methods on the small new domains. INFS achieves mean values of 66.06%, 56.66%, and 62.11 in OA, AA, and Kappa on the KSC dataset. All training adapters on top of base model methods outperform INFS in all three metrics with an obvious improvement. Note that TA-BN-1 and TA-BN-2 reuse all convolution layers from the base model and only adapt the parameters in BN layers. This proves that the base model has learned important common knowledge from the five preexisting datasets, which boosts the classification accuracy of two new small datasets. Comparing the results of models trained from base1 and base2, we find that TA-BN-2 and TA-BNPW-2 achieve better classification performance consistently than their counterparts trained from base1. The mean values of OA, AA, and Kappa of TA-BN-2 reach 81.44%, 75.40%, and 79.34 on the KSC dataset, respectively. In this setting, more preexisting data are used to train the base model, and the performance on new datasets has also increased. The reason

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could be that a better base model contains more common knowledge useful for the classification of the new domains. The results on the Botswana dataset show similar trends with the highest mean value of metrics as 91.40% OA, 92.30% AA, and 90.68 Kappa, significantly outperforming INFS. On the Botswana dataset, the train adapter methods also demonstrate the performance of lower standard deviation compared to INFS. Overall, the results demonstrate that our multidomain design can learn useful knowledge from preexisting domains and is easy to generalize to new domains with very few training samples.

V. Conclusion

The traditional paradigm of deep learning-based HSI classification usually trains an individual model for each data domain. This is expensive and inefficient. To overcome this drawback, we propose a universal network with domain adapters to learn universal representations for MHIC. The proposed adapters can adapt the features in a base network by adjusting the distribution of features according to each domain of data. Specifically, the base network learns the common knowledge of all datasets. The domain-specific BN helps the MDN to learn to normalize individual datasets according to the statistical distribution. The PW convolution adapters further allow a small number of parameters to address domain-specific properties. We conducted comprehensive experiments to validate the usefulness of the MDNs and universal representations in the context of MHIC. The results show that our proposed networks can address this challenging classification problem with better or comparable accuracy as traditional individual networks while significantly reducing model sizes.

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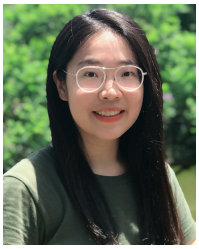
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